

Analysis PhD Exam, August 2007

Attempt *EIGHT* questions. To receive credit be sure to state clearly the theorems used in solutions.

1. Let $(X, \mathcal{S}, \mu)$ be the following measure space: X is the set of positive integers, $\mathcal{S}$ is the power set of X , and μ is the counting measure. If f is μ -integrable, does $\sum |f(m)|$ converge? Prove. Is the converse true? Prove.
2. Let μ and ν be finite valued signed measures such that $\mu \ll \nu$ and $\nu \ll \mu$. Prove that $\left(\frac{d\mu}{d\nu}\right)\left(\frac{d\nu}{d\mu}\right) = 1$ a.e.
3. Let Σ be an algebra of subsets of a set S and $\sigma(\Sigma)$ the σ -algebra generated by Σ . State the Hahn extension theorem for a σ -finite measure μ defined on Σ . Use a Monotone Class argument to prove that the extension is unique.
4. Let $\{x_m\}$ be a sequence of elements from a Banach space $\mathcal{X}$. Suppose $\sum \|x_m\| < \infty$. Does $\sum x_m$ converge unconditionally? Prove.
5. Let $\mathcal{M}$ and $\mathcal{N}$ be closed linear subspaces of a Hilbert space $\mathcal{H}$. Assume $\mathcal{M} \perp \mathcal{N}$. Show $\mathcal{M} + \mathcal{N}$ is closed.
6. Let $\mathcal{X}$ be a Banach space and suppose $\Pi : \mathcal{X} \rightarrow \mathcal{X}^{**}$ is the natural embedding map. Is $\Pi(\mathcal{X})$ closed in $\mathcal{X}^{**}$? Prove.
7. Let $L^1(\mathbb{R})$ be the space of Lebesgue integrable functions on $\mathbb{R}$. Assume E is a Lebesgue measurable set. Denote by Y the set of all functions $f \in L^1(\mathbb{R})$ such that $f(t) = f(-t)$ for almost all $t \in E$. Prove Y is closed in $L^1(\mathbb{R})$.
8. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{D}$ a sub- σ -algebra of $\mathcal{F}$. Suppose $f \in L^2(\Omega, \mathcal{F}, P)$. Show by Hilbert space theory that there exist a $\mathcal{D}$ -measurable function h such that

$$\int_D f dP = \int_D h dP, \quad D \in \mathcal{D}.$$

9. Let $\mathcal{Y}$ and $\mathcal{Z}$ be closed linear subspaces of the Banach space $\mathcal{X}$ such that each $x \in \mathcal{X}$ has a unique representation $x = y + z$, where $y \in \mathcal{Y}$ and $z \in \mathcal{Z}$. Show that there exists a constant K such that $\|y\| \leq K\|x\|$ and $\|z\| \leq K\|x\|$, for all $x \in \mathcal{X}$, where $x = y + z$ is the above representation of x .